OcuSmart: AI-Powered Portable Autorefractor

Capstone Project Report

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ABSTRACT

One of the main causes of visual impairment in the world is refractive errors, especially in rural and low-resource areas where access to ophthalmic diagnostic equipment is restricted. The design and development of an inexpensive, portable, artificial intelligence (AI)-based autorefractor for objective eye screening is presented in this paper. To estimate refractive errors like myopia, hyperopia, astigmatism and colour blindness, the suggested system combines infrared illumination, optical sensing hardware, and an AI-driven software module. The gadget is ideal for screening and primary healthcare applications because it is small, inexpensive, and simple to use. Results from experiments show that the suggested system can produce accurate refractive measurements, providing a viable substitute for traditional autorefractors.

INTRODUCTION

Vision is one of the most critical senses for human interaction with the world. According to the World Health Organization (WHO), a major chunk of the population globally suffer from vision impairment or blindness, and a large part of these cases are preventable or remain unaddressed due to lack of timely diagnosis and affordable treatment options. Among these visual impairments, refractive errors—including myopia (near-sightedness), hyperopia (farsightedness), and astigmatism—are the most common and easily correctable causes. These errors occur when the eye's optical system is unable to focus light accurately on the retina, resulting in blurred vision and reduced visual acuity. The impact of uncorrected refractive errors goes beyond vision loss. Studies have shown that poor eyesight can lead to decreased academic performance in children, reduced work productivity in adults, and a decline in quality of life, particularly for elderly individuals. In developing nations and rural areas, where access to optometric help is limited, these conditions often go undiagnosed, leading to lifelong challenges for affected individuals.

A major bottleneck right now is how traditional eye screenings are done. They rely on heavy, expensive table-mounted equipment and require a trained professional to operate and interpret the results. Furthermore, standard checkups often require mydriatic eye drops to forcefully dilate the patient's pupils. This is physically uncomfortable, takes too much time, and leaves the patient's vision blurry for hours. Because of this, traditional methods are highly impractical for mass screening camps. This is where digital health and low-cost engineering come into play. With the recent drop in the cost of off-the-shelf hardware like high-resolution CMOS cameras and microprocessors, we have a clear opportunity to build accessible, handheld diagnostic tools. At the same time, artificial intelligence—particularly deep learning using Convolutional Neural Networks —has shown remarkable accuracy in medical image analysis, such as detecting retinal diseases and glaucoma. By integrating these AI algorithms with portable camera modules, we can bypass the need for expensive clinical setups and dilating drops, making eye screening faster, cheaper, and fully automated.

This project aims to bridge the gap between clinical accuracy and everyday accessibility, offering a solution that could revolutionize teleophthalmology and promote equitable eye care worldwide. By reducing the dependence on physical infrastructure and expert intervention, the proposed system can play a pivotal role in expanding access to vision health services across geographically and economically challenged regions

NEED ANALYSIS

The Clinical Gap: Inefficiencies in Traditional Eye Screening

Current methods for detecting visual impairments rely heavily on traditional clinical settings, which present several significant bottlenecks.

- Traditional eye screenings are time-consuming procedures.
- These procedures require the presence of professionals to administer and interpret the results.
- Standard practices often involve the use of mydriatic eye drops to dilate the pupil, which can cause physical discomfort for the patient.

The Accessibility Gap: The Need for Mass Screening

Because current diagnostic methods are slow and require clinical infrastructure, widespread preventative care is limited.

- There is a critical need for an affordable solution designed specifically for mass vision screening.
- To bridge the accessibility gap, the solution must be a portable, non-invasive device that simplifies the screening process and makes it accessible to a broader population.
- Efficiency is necessary as screening needs to yield results in seconds rather than minutes.

The Pediatric and Non-Verbal Patient Gap

Standard visual acuity tests and color blindness screenings often require subjective verbal feedback from the patient.

- This reliance on patient communication creates a barrier for infants and non-verbal individuals for the diagnosis.
- There is a distinct need for an objective screening method, such as utilizing chromatic pupillometry to measure the pupil's constriction response, to accurately assess these demographics.

The Diagnostic Scope Requirement

A viable mass-screening tool must be capable of identifying the most prevalent ocular issues simultaneously.

- The device must implement real-time algorithms to identify common refractive errors, such as Myopia, Hyperopia, and Astigmatism.
- Beyond standard refractive errors, the device should address ocular health by incorporating screening methods for Color Vision Deficiency (CVD) and overall retinal health.

OBJECTIVES AND DELIVERABLES

Objectives with which the project will be carried out :

- Objective 1: To design and prototype a handheld, non-mydriatic photo-screening device that provides an affordable solution for mass vision screening.
- Objective 2: To implement real-time algorithms that identify common vision defects, specifically Myopia, Hyperopia, and Astigmatism.
- Objective 3: To develop a screening method for Color Vision Deficiency (CVD) by measuring the pupil's constriction response to different visible light spectrums.
- Objective 4: To establish a method for assessing overall retinal health utilizing the same pupillary constriction response mechanism.

Deliverables that will be obtained:

1. Hardware Deliverables
   - OcuSmart Prototype: A handheld unit containing a Raspberry Pi, Sony Global Shutter camera, 850nm IR LEDs, and a 3.5-inch touchscreen.
   - Optical Assembly: A custom-aligned 25mm lens and IR filter setup for high-resolution retinal reflection capture.
2. Software Deliverables
   - Vision Suite: Python/OpenCV scripts for real-time pupil detection and red reflex geometric analysis.
   - Diagnostic Engine: An algorithm that classifies refractive errors and provides a "Pass/Refer" status.
   - Chromatic Module: Software control for red/blue light pupillometry to test retinal sensitivity.
3. Design Goals
   - Portability: Handheld and battery-operated for field use.
   - Non-Mydriatic: Not an invasive procedure.
   - Cost-Effectiveness: Total build cost under ₹25,000

COST ANALYSIS

Component	**Quantity**	**Cost (₹)**
Raspberry Pi 4 (4GB)	1	6,958
Camera Module (IMX296)	1	8,000
SanDisk Ultra 32GB MicroSD	1	700
100-Ohm (¼ Watt) Resistors	5	706
IRF520 MOSFET Driver Module	2	74
18650 Lithium-Ion Batteries	2	160
Official Pi 15W Power Supply	1	691
VL53L1X ToF Sensor	1	582
Total Cost		**17,871**

PROJECT SCOPE AND DESIGN GOALS

PROJECT SCOPE

The scope of the OcuSmart project comprises the development of a fully functional, AI-powered diagnostic hardware prototype and its underlying software algorithms. Specifically, the project contains:

- **Device Engineering:**
  Building a handheld, portable autorefractor using commercially available, cost-effective components, including a Raspberry Pi 4 compute unit, a global shutter camera (SONY IMX296), a Time-of-Flight (ToF) distance sensor, and custom illumination arrays (IR, Red, and Blue LEDs).

- **Refractive Error Detection:**
  Implementing computer vision algorithms (via OpenCV) to capture the eye's "red reflex" using near-infrared (850nm) illumination. The system will analyze the geometry of the "crescent moon" light reflection and the Hirschberg Ratio to classify common refractive errors: Myopia, Hyperopia, and Astigmatism, as well as ocular misalignment.

- **Chromatic Pupillometry Integration:**
  Developing a secondary screening module that uses visible light spectrums (660nm Red and 470nm Blue LEDs) to measure dynamic pupil constriction frame-by-frame. This will serve as an objective screening method for Color Vision Deficiency (CVD), such as Deuteranopia/Protanopia, and overall retinal health.

- **Automated Diagnostics:**
  Creating an onboard processing pipeline that interprets the gathered optical data in real-time to render an immediate, automated "Pass" or "Refer" (or "Healthy" vs. specific condition) result on an integrated touchscreen interface.

DESIGN GOALS

The engineering and user-experience decisions for OcuSmart are driven by the need to overcome the limitations of traditional optometry. The core design goals are:

- **Portability and Accessibility:**
  To transition vision screening from static, specialized clinical environments to a scalable, handheld form factor capable of mass screening in rural or resource-limited areas.

- **Non-Invasive Comfort (Non-Mydriatic):**
  To eliminate the need for uncomfortable mydriatic eye drops (which force pupil dilation), making the screening process significantly more comfortable and patient-friendly.

- **Speed and Autonomy:**
  To replace time-consuming traditional procedures with a system that works in seconds and reduces the strict dependency on highly trained medical professionals to interpret the results.

- **Objective Assessment:**
  To provide accurate diagnostic capabilities without requiring subjective verbal feedback from the patient, making it highly effective for infants and non-verbal individuals.

- **Economic Viability:**
  To maintain an affordable production cost by implementing a component-driven architecture, with a target build cost of approximately ₹18,000 to ₹25,000 INR, ensuring the solution remains financially accessible for widespread deployment.

LITERATURE REVIEW

Artificial intelligence has become central to ophthalmic diagnostics over the past decade, with deep learning increasingly used to analyze retinal and anterior-segment imagery as part of a broader push toward automated screening in settings with few trained specialists [1], [2].

Convolutional neural networks (CNNs) form the basis of most of this work. Abramoff et al. demonstrated autonomous diabetic retinopathy detection [3], Asaoka et al. applied deep learning to glaucoma classification from standard perimetry [4], Burlina et al. graded age-related macular degeneration from fundus images [5], and Gulshan et al. and Li et al. reported diabetic retinopathy detection directly from fundus photographs [6], [7], building on the deep learning paradigm established by LeCun, Bengio, and Hinton [8]. De Fauw et al. extended this to a combined diagnosis-and-referral pipeline spanning multiple retinal conditions [9] — the same kind of automated screen-then-refer decision OcuSmart's diagnostic engine is intended to produce, for a different clinical target.

Cost and portability, more than raw accuracy, remain the harder problem. Choi et al. describe a smartphone-based portable fundus camera [10], and a more direct precedent is the AI-powered portable retinal health assessment device reported by Sakthi Kaviya et al. [11], whose low-cost, field-deployable goals parallel this project's, though it addresses general retinal health rather than non-mydriatic refractive-error screening specifically.

Preprocessing quality affects everything downstream. Gaussian and median filtering, histogram equalization, and CLAHE are established methods for noise suppression and contrast normalization ahead of feature extraction [12], [13], and the same operations form Module 2 of OcuSmart's software pipeline.

Optical Coherence Tomography, first demonstrated by Huang et al. [14], remains the clinical gold standard for cross-sectional retinal imaging, but its cost, bench-mounted form factor, and dependence on trained operators exclude it from mass screening — the gap portable alternatives are trying to close. Teleophthalmology partially addresses this by separating capture from diagnosis, with systematic reviews reporting improved reach into underserved regions when portable capture is paired with remote interpretation [15], [2] — the same local-capture, remote-or-automated-inference split OcuSmart adopts, minus the connectivity dependence.

Photorefraction and the Hirschberg ratio. Non-mydriatic refractive-error screening has an established basis in photorefraction, first formalized by Howland and Howland [16], where infrared illumination produces a red reflex whose geometry varies systematically with refractive state. The Hirschberg ratio — reflex displacement over pupil radius, $H = d/r$ — is a long-established clinical measure originally used for strabismus angle estimation; Brodie's photographic calibration of the test [17] established the millimeter-to-angle conversion that later photorefraction-based adaptations, including this project's, build on. This is the same measurement OcuSmart's computation engine uses as its core feature, and its clinical precedent is what justifies treating H as a physically meaningful proxy for refractive error rather than an arbitrary image statistic.

Chromatic pupillometry — tracking pupillary constriction under red and blue stimuli — has similarly been used to probe photoreceptor health, colour vision deficiency, and retinal function [18], forming the basis of OcuSmart's secondary screening module.

Global shutter versus rolling shutter sensors. Most low-cost embedded cameras use rolling-shutter sensors, exposing each row sequentially, which makes them prone to motion artifacts when imaging fast-moving subjects [12], [13]. Given that the pupillary light reflex occurs on a short timescale, this project treats that susceptibility as a material design risk rather than a cosmetic one — an engineering judgment specific to this application rather than a claim drawn from the ophthalmic literature itself. Global-shutter sensors, exposing all pixels simultaneously, are the standard machine-vision mitigation for rapid motion [12], which directly motivates OcuSmart's choice of the Sony IMX296 over the rolling-shutter modules typical of low-cost camera hardware.

Transfer learning for on-device medical classification. Training CNNs from scratch requires large labeled datasets rarely available to a single-institution project [8]. Transfer learning — fine-tuning a network pretrained on a large general dataset — is well established as the standard response to this constraint, with Kim et al.'s review of 121 medical-imaging studies confirming its efficacy despite data scarcity [19]. Lightweight architectures such as MobileNetV2, introduced by Sandler et al. [20], are specifically designed for power- and compute-constrained embedded hardware, which is the basis for running a MobileNetV2 backbone on a Raspberry Pi rather than a cloud-hosted network.

Taken together, the literature shows two maturities that have not yet been combined in one low-cost device: accurate but infrastructure-heavy AI diagnostic models, and portable capture hardware that has not generally paired non-mydriatic photorefraction with on-device AI classification. Portable devices such as [11] address general retinal health; teleophthalmology [15] addresses remote interpretation but assumes connectivity; OCT [14] delivers precision at the cost of portability and specialist dependence. None combines non-mydriatic operation, sub-two-second on-device response, and a bill-of-materials cost low enough for mass deployment by non-specialist operators. OcuSmart is positioned in this gap: a global-shutter IR imaging system paired with a Hirschberg-ratio photorefraction engine and a lightweight on-device MobileNetV2 classifier, addressing the portability, cost, and usability constraints the literature identifies but does not, together, resolve.

ENGINEERING DESIGN FEATURE & SPECIFICATIONS

DESIGN FEATURES:

- **Portable and Compact:**
  The device should be lightweight and handheld, making it easy to use in clinics and field environments.
- **User-Friendly and Non-Invasive:**
  It must operate without eye drops and be simple enough for use by non-experts.
- **Accurate and Fast Results:**
  The system should detect refractive errors using AI and provide quick, reliable outputs.
- **Efficient Imaging System:**
  It should include proper camera, IR illumination, and sensors to capture clear eye images.
- **Low Cost and Power Efficient:**
  The device should be affordable and energy-efficient for practical and wide-scale usage.

DESIGN SPECIFICATIONS:

1. **Raspberry Pi 4 - 4GB (Processing Unit):**
   - **Role:** Acts as the central computing unit of the device.
   - **Implementation**: It houses the onboard processing pipeline, which runs the Python/OpenCV Vision Suite for real-time pupil detection and the diagnostic engine. It executes the real-time algorithms required to analyze optical data and classify refractive errors (Myopia, Hyperopia, Astigmatism) without relying on an external computer.

2. **SONY IMX296 (Camera):**
   - **Role:** Captures the required high-resolution images of the eye.
   - **Implementation:** This specific global shutter camera is utilized to prevent motion blur when capturing the fast pupillary responses and the retinal "red reflex phenomenon".

3. **25mm M12 Lens (Lens):**
   - **Role:** Focuses the incoming light onto the camera sensor.
   - **Implementation:** It forms part of a custom-aligned optical assembly designed specifically to capture high-resolution retinal reflections clearly.
4. **850nm IR Filter (Filter):**
   - **Role:** Blocks ambient visible light and only allows near-infrared light to pass to the sensor.
   - **Implementation:** Paired with the lens, it isolates the specific 850nm wavelength used to illuminate the eye, ensuring the computer vision algorithms capture a clear "crescent moon" light reflection for geometric analysis.
5. **IR LEDs:**
   - **Role:** Safely illuminates the interior of the eye.
   - **Implementation:** The 850nm IR LEDs provide the necessary light to capture the eye's "red reflex". Because infrared light is invisible to the human eye, it acts as a non-mydriatic solution, meaning it does not cause the pupil to constrict or require the use of uncomfortable dilating eye drops.
6. **MOSFET:**
   - **Role:** Acts as a high-speed electronic switch to drive the LED arrays.
   - **Implementation:** The Raspberry Pi’s standard GPIO pins cannot source enough electrical current to directly power the custom illumination arrays (the 850nm IR, 660nm Red, and 470nm Blue LEDs). The MOSFET solves this by taking a low-power control signal from the Raspberry Pi to rapidly switch on a separate, higher-current power supply from the batteries. This allows the system to trigger short, intense bursts of light that are perfectly synchronized with the SONY IMX296 global shutter camera, capturing the eye's red reflex frame-by-frame without draining the compute unit or burning out the pins.
7. **VL53L1X ToF (Distance Sensor):**
   - **Role:** Measures the physical distance between the device and the patient's eye.
   - **Implementation:** This Time-of-Flight (ToF) distance sensor helps ensure the camera is positioned at the exact focal distance required for an accurate, objective reading.
8. **Display Touchscreen (User Interface):**
   - **Role:** Displays real-time information to the operator.
   - **Implementation:** A 3.5-inch touchscreen integrated into the unit renders immediate, automated diagnostic results, showing a clear "Pass" or "Refer" status, making the device accessible and user-friendly for non-expert operators.

9. Batteries (Power Supply):

- **Role:** Provides untethered electrical power.
- **Implementation:** Ensures the device is fully battery-operated, satisfying the design goal of being a handheld unit suitable for field environments and mass screening in resource-limited areas.

10. Case (Enclosure):

- **Role:** Protects the internal hardware and provides the physical structure.
- **Implementation:** Houses the custom-aligned optical assembly, computing unit, and sensors in a lightweight, compact form factor designed for comfortable handheld clinical use.

HARDWARE DESIGN METHODOLOGY

The hardware architecture of OcuSmart is built around four constraints: non-mydriatic operation (no dilating drops), a sub second response time, objective measurement independent of patient feedback, and a total bill-of-materials cost under ₹25,000. Rather than adapting an existing clinical autorefractor design, each subsystem below was selected component-by-component against these constraints, to keep both cost and development risk low. The subsystems are presented in the order signal and power actually flow through the device: optical capture, illumination, ranging, compute, and mechanical integration.

1. Optical Subsystem

Sensor: Sony IMX296

The core decision is global shutter versus rolling shutter. A rolling shutter sensor exposes each row of pixels sequentially, over a span of several milliseconds. The pupillary light reflex is fast enough that a rolling shutter will capture the top and bottom of a frame at meaningfully different points in the pupil's response, smearing the crescent-shaped red reflex and corrupting the Hirschberg ratio calculation downstream. The IMX296 avoids this as every pixel is exposed and read out at the same instant.

Key specifications relevant to the design:

- 1.6 MP resolution (1456 × 1088 active pixels), 1/2.9" optical format
- 3.45 µm × 3.45 µm pixel pitch — large for this resolution class, keeping light sensitivity high even though the IR illumination pulse is brief
- Native frame rates up to 60 fps at full resolution — comfortably above the 15–20 fps pipeline target, leaving headroom for processing rather than running the sensor at its ceiling

Lens: 25mm M12

A fixed 25mm M12-mount lens was chosen over a variable-focus lens for two reasons that matter for a field device: fewer moving parts to misalign during handheld use, and a predictable, calibratable field of view. The working distance this focal length produces at the eye is the same distance the VL53L1X ranging subsystem (Section 3) is tuned to hold constant.

Filter: 850nm IR Bandpass

This narrowband optical filter sits in front of the sensor and passes only wavelengths around 850nm i.e. the illumination wavelength blocking the visible light. Without it, ambient room light would flood the sensor with visible-spectrum noise and wash out the comparatively faint IR red-reflex signal. The filter is what makes the pipeline usable outside a darkened clinical room, which is a hard requirement for mass screening.

2. Illumination Subsystem

Two Independent LED Banks

Bank	Wavelength	Function
Bank A	850nm (IR)	Non-mydriatic red-reflex capture for refractive-error screening
Bank B	660nm (red) / 470nm (blue)	Chromatic pupillometry for colour vision deficiency and retinal-health screening

The 850nm choice for Bank A is deliberate on two fronts: it sits outside the visible spectrum i.e. 400–700nm, so the pupil does not constrict in response to it and it still falls within the spectral response range of the IMX296's silicon sensor, so the sensor can register what the human eye cannot see.

MOSFET Drive Stage

The Raspberry Pi's GPIO pins can safely source only a small current per pin, well under what LED arrays bright enough for reliable reflex capture require. Driving the LEDs directly from GPIO would either fail to produce usable illumination or risk damaging the Pi. The IRF520 N-channel power MOSFET solves this by acting as a switch: GPIO controls the gate (low-current signal), while the LED array's actual operating current is drawn from the battery rail through the MOSFET's drain-source channel.

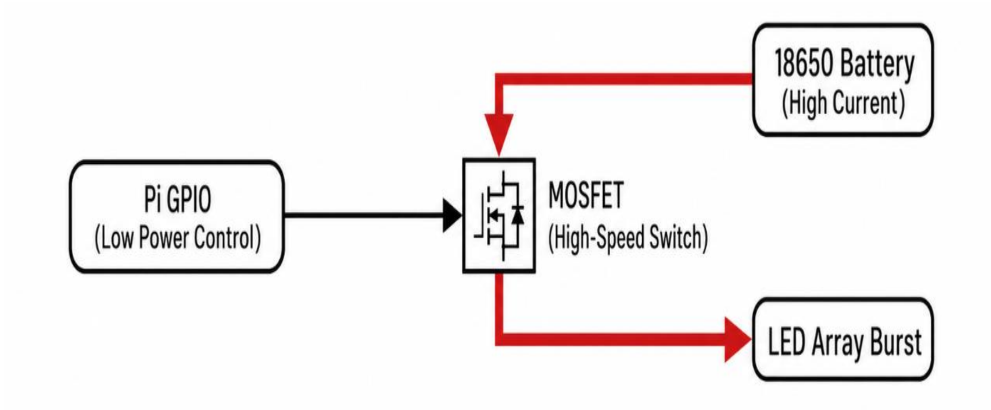

Figure 1: Control Logic for MOSFET Drive Stage

Characteristics of the components chosen:

- Gate threshold voltage: 2–4V: Pi's 3.3V GPIO logic level is sufficient to begin turning the MOSFET on.
- Continuous drain current rating: 9.2A at 25°C: The available capacity significantly exceeds the power requirements of the LED arrays which provides a substantial operating margin.
- On-state resistance 0.27Ω: This temperature is low enough to keep self-heating manageable at these currents.
- Drain-source breakdown voltage: 100V: This voltage is well above the 7.4V dual-18650 rail which implements that there is no risk of breakdown.

Synchronization

The GPIO pin driving the MOSFET gate is timed to pulse within the same window as the camera's exposure trigger.

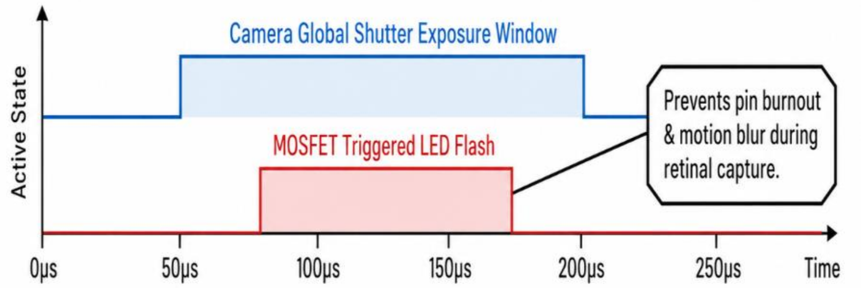

Figure 2 :Timing Waveform for the Synchronization of Camera and MOSFET

3. Ranging Subsystem

VL53L1X Time-of-Flight Sensor

This sensor measures distance by timing how long an emitted pulse of light takes to reflect off a target and return. Specifications relevant to this application:

- Ranging distance: 4 cm to 400 cm
- Accuracy: ±25mm (±20mm in low ambient light)
- Update rate: up to 50 Hz which is fast enough to track head position in real time without perceptible lag
- Emitter: 940nm Class 1 laser, invisible and eye-safe by design which is going to be relevant given the sensor's proximity to the patient's eye
- Interface: I2C at address 0x29, sharing the same two-wire bus (SDA/SCL) used elsewhere on the Pi's GPIO header, keeping wiring simple

4. Compute and Interface Subsystem

Raspberry Pi 4 (Model 4(b) 4GB)

It is chosen to run the entire OpenCV pipeline on-device rather than requiring a paired laptop or cloud connection which is the single biggest factor in making the unit usable in a screening camp with no reliable internet. The 4GB RAM variant was chosen to give the pipeline, frame buffer, and GUI layer headroom without triggering swap, which would introduce latency spikes inconsistent with the sub second response target.

Interface	Connected Component	Function
CSI-2 connector	IMX296 camera module	Dedicated high-bandwidth image data path
I2C bus (GPIO 2/3)	VL53L1X ToF sensor	Distance feedback for focal positioning
GPIO output pin	IRF520 MOSFET gate	LED bank switching, synced to camera trigger
SPI / DSI	3.5" touchscreen	Operator display and Pass/Refer output

Touchscreen

Component is chosen because it renders the full result set — Pass/Refer plus Sph/Cyl/Axis values without needing a paired phone or laptop app.

5. Power Subsystem

Dual 18650 Li-ion Cells

18650-format lithium-ion cells were chosen specifically because they are a widely available, standardized form factor with well-characterized capacity and discharge behaviour. The two cells feed a regulated rail that steps down to the 5V/3A the Raspberry Pi 4 requires, with a separate tap or regulator stage for the LED drive rail through the MOSFET.This is what makes the unit field-deployable.

6. Mechanical Integration

The enclosure has one job beyond simply housing components: it must hold the LED–eye–lens–sensor optical path in a fixed geometric relationship, and it must block ambient IR from reaching the sensor from any path other than the reflex off the patient's eye. Any flex or misalignment in this path directly degrades the accuracy of the Hirschberg ratio calculation, since that calculation depends on precise geometric measurement of reflex displacement relative to pupil centre.

SOFTWARE METHODOLOGY

System Overview

The proposed system architecture is organized into eight functional modules, structured across two operational layers. Modules 1 through 6 constitute the deployed runtime pipeline which consist of the sequence of operations executed in real time on the embedded platform during an active patient screening. Modules 7 and 8 constitute the offline development and validation framework, encompassing the testing infrastructure used prior to and during system development rather than during a live screening session. This distinction is maintained throughout the following sections, as the two layers serve fundamentally different purposes within the overall system lifecycle.

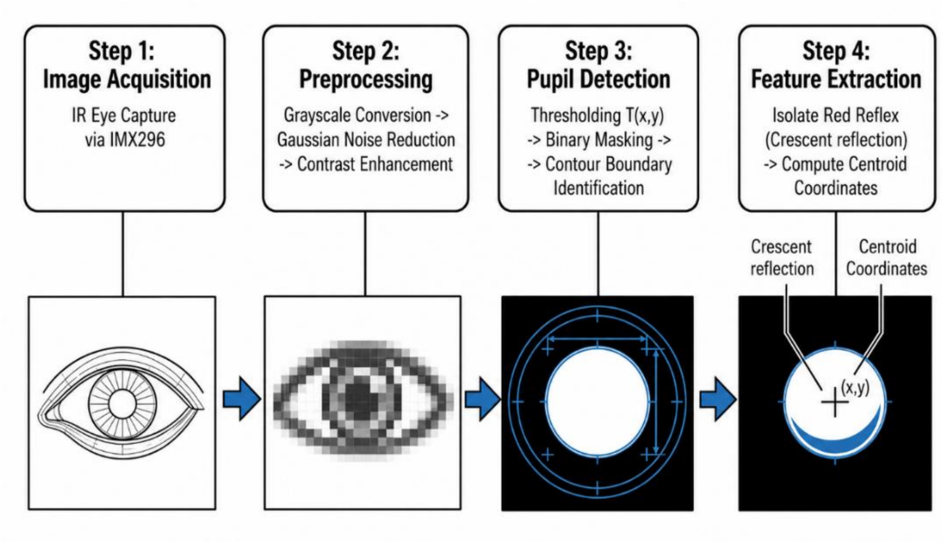

Figure 3: Flow Diagram of the detection process

1. Image Acquisition

It constitutes the physical data-capture layer and interfaces directly with the hardware subsystems described in the hardware design methodology. It functions as the software abstraction layer governing the hardware components previously documented, without introducing additional processing logic of its own.

- Camera Interface provides the software-level driver and API layer responsible for communication with the image sensor over the MIPI CSI-2 bus, managing exposure control, gain settings, and frame timing.
- IR LED Control submodule governs the GPIO-driven MOSFET trigger circuit, synchronizing the 850nm illumination pulse with the sensor's exposure window to ensure each captured frame corresponds to a known, controlled illumination event.

- Frame Capture function performs the acquisition and buffering of the image data, initiated by the capture command issued from the operator interface.
- ToF Distance Measurement submodule continuously polls the Time-of-Flight sensor over the I2C bus, gating the capture process such that image acquisition proceeds only once the patient's eye is confirmed to lie within the optical system's effective focal range.

2. AI Software Pipeline

It performs deterministic computer-vision processing on the acquired frame, preparing the image data for the geometric computations carried out in Module 3. Despite its designation, this module does not employ a trained machine-learning model; all operations are classical image-processing techniques.

Image pre-processing converts the captured frame to grayscale and restricts processing to the relevant region of interest. Noise removal is achieved through Gaussian filtering. Contrast enhancement employs Contrast Limited Adaptive Histogram Equalization (CLAHE) to normalize local contrast variation arising from differences in skin tone, iris pigmentation, and ambient lighting conditions. Eye detection performs coarse localization of the eye region within the broader frame prior to pupil-level analysis. Pupil segmentation isolates the pupil boundary using the Hough Circle Transform, exploiting the near-circular geometry of the pupil. Corneal reflection detection identifies the bright infrared reflex within the segmented pupil region through intensity-based blob detection, yielding the displacement value subsequently used in the Hirschberg ratio calculation. Feature extraction consolidates the pupil centre, pupil radius, and reflex position into numerical outputs for downstream processing.

3. Mathematical Computation Engine

It constitutes the core optical and geometric computation stage of the system, converting pixel-domain measurements into physically meaningful, clinically interpretable values.

- Hirschberg Ratio Calculation computes $H = d/r$, where d denotes the corneal reflex displacement and r denotes the pupil radius.
- Crescent Area Estimation quantifies the spatial extent of the visible red-reflex crescent, a secondary indicator correlated with the magnitude of refractive error.
- Pixel-to-Millimetre Conversion establishes the calibration between image-domain pixel measurements and real-world physical dimensions, using the known focal length, sensor pixel pitch, and the working distance confirmed by the Time-of-Flight sensor; this conversion underlies the physical validity of all subsequent measurements.

- Optical Geometry computation applies similar-triangles and thin-lens relationships to reconstruct real-world ocular geometry from the two-dimensional image data.
- Refractive Power Estimation converts the computed Hirschberg ratio into an estimated dioptric value, using an empirically derived conversion constant drawn from the photorefraction literature; the specific constant employed should be explicitly cited in the final documentation.
- Statistical Error Correction applies multi-frame averaging and outlier rejection (such as blinks or motion) to stabilize the measurement prior to its use in AI Diagnostic Engine.

4. AI Diagnostic Engine

- Feature Vector Generation assembles the outputs of Module 3 (Hirschberg ratio, crescent area, refractive power estimate, and reflex displacement) into a fixed-length numerical input vector.

- AI Model Prediction is implemented using a lightweight classical machine-learning model (e.g., logistic regression, support vector machine, or random forest) rather than a convolutional architecture; the specific model type used should be stated explicitly in the final documentation.

The model outputs predictions for Myopia, Hyperopia, and Astigmatism, accompanied by a Confidence Score representing the per-class prediction probability, which informs the severity-thresholding logic of decision & diagnostic logic.

5. Decision & Diagnostic Logic

It translates the raw output of Module 4 into a clinically actionable recommendation. Severity Classification grades the estimated dioptric value into defined severity bands (e.g., mild, moderate, severe). Clinical Threshold Checking compares the estimated refractive error against standard screening cutoffs established in the relevant clinical literature. Diagnostic Recommendation generates the binary Pass/Refer output constituting the system's primary screening decision. Report Generation compiles the structured diagnostic result set for presentation via Operator User Interface.

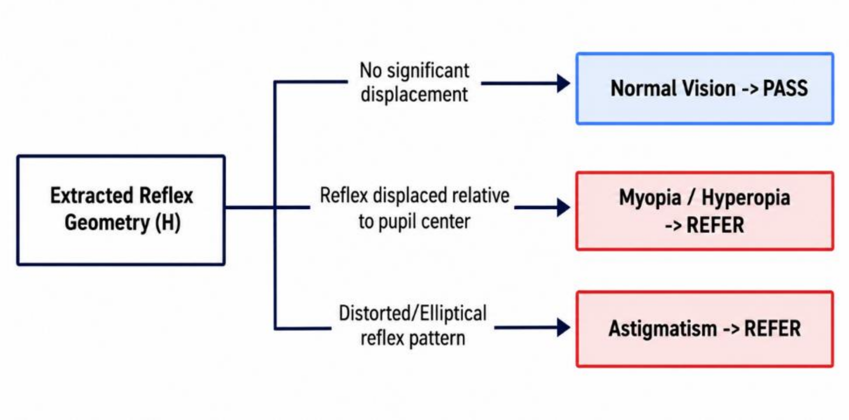

Figure 4 : Diagnostic Flow Diagram

6. Operator User Interface

It implements the operator-facing interface rendered on the system's touchscreen display. Patient Registration provides intake and local record creation. Live Camera Feed presents a real-time preview enabling operator alignment, used in conjunction with the distance feedback from Module 1. The Capture Button initiates frame acquisition. Results Display presents the Pass/Refer outcome together with the estimated spherical, cylindrical, and axis values. Historical Records provides local storage and retrieval of prior screening results, a function of particular importance in mass-screening deployments processing multiple patients per session.

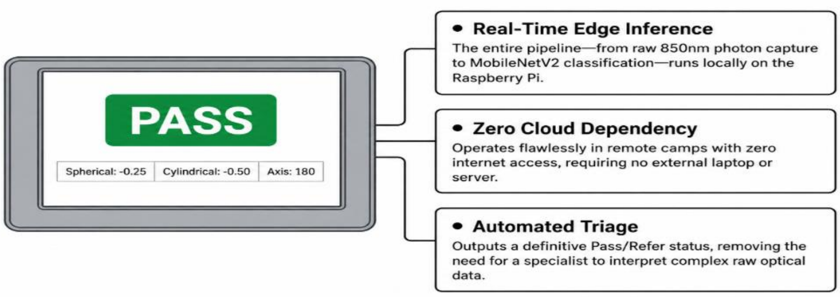

Figure 5 : User Interface

7. Simulation & Validation

Virtual Eye Dataset provides synthetic or simulated eye imagery for testing prior to clinical deployment, enabling coverage of edge cases not readily available through clinical sourcing. Algorithm Testing comprises unit and integration testing of the computer-vision and mathematical pipeline in isolation. MATLAB/Python Simulation provides bench-level modelling of the optical and signal-processing pipeline independent of the embedded implementation. Hardware-in-the-Loop Testing evaluates the physical device against controlled optical targets of known, calibrated refractive properties.

8. Performance Metrics

- Classification metrics such as Accuracy, Precision, Recall, F1 Score, Sensitivity, Specificity, and ROC-AUC evaluate the diagnostic classification performed by Module 4, with Sensitivity/Recall assigned particular weight given that missed detection of a genuine refractive or pathological condition carries greater clinical consequence than a false positive referral.
- Regression metrics (Mean Absolute Error (MAE) and Root Mean Square Error (RMSE)) evaluate the continuous dioptric estimate produced by Module 3 against reference autorefractor measurements.
- System metrics which influence the inference time, processing latency, frame rate, and memory usage help in validating the system's real-time performance claims against the operational targets established in the project's need analysis

AI MODEL METHODOLOGY

Dataset Acquisition

The AI model was developed using the Myopia Image Dataset obtained from Kaggle. The dataset consists of retinal and eye images categorized into two classes: Myopia and Normal. The dataset was downloaded programmatically using the kagglehub library and organized into separate folders for each class.

Data Preprocessing

Before training, the images underwent preprocessing to ensure consistency and improve model performance.

The preprocessing steps included:

- Loading image paths and corresponding labels into a Pandas DataFrame.
- Assigning numerical labels:
  - Myopia = 1
  - Normal = 0
- Splitting the dataset into 80% training data and 20% validation data using train_test_split.
- Resizing all images to 128 × 128 pixels.
- Normalizing pixel values from the range 0–255 to 0–1 using ImageDataGenerator.

These preprocessing steps reduce computational complexity while providing standardized inputs for the convolutional neural network.

Convolutional Neural Network (CNN) Architecture

A Convolutional Neural Network (CNN) was implemented using TensorFlow and Keras to automatically extract visual features associated with myopic and normal eye images.

The architecture consists of:

- Three Convolutional layers with 32, 64, and 128 filters respectively.
- Three Max-Pooling layers to reduce spatial dimensions and computational complexity.
- A Flatten layer to convert extracted feature maps into a one-dimensional feature vector.
- A Fully Connected Dense layer with 128 neurons using the ReLU activation function.
- A Dropout layer with a dropout rate of 0.3 to reduce overfitting.
- A final Dense layer with a Sigmoid activation function for binary classification.

The CNN automatically learns hierarchical image features such as edges, textures, and complex retinal patterns without requiring manual feature engineering.

Model Training

The model was trained using the Adam optimizer, which provides adaptive learning rates for efficient convergence.

The following training parameters were used:

During training, the validation dataset was used to monitor model performance after each epoch.

Parameter	Value
Optimizer	Adam
Loss Function	Binary Crossentropy
Evaluation Metric	Accuracy
Batch Size	32
Epochs	10

Model Checkpointing

To preserve the best-performing model, the ModelCheckpoint callback was implemented. The model with the highest validation accuracy was automatically saved during training. Additionally, a CSVLogger recorded the training and validation metrics after each epoch for further analysis.

Performance Evaluation

The trained model's performance was evaluated by monitoring:

- Training Accuracy
- Validation Accuracy
- Binary Crossentropy Loss

Accuracy graphs were generated using Matplotlib to visualize the learning behavior of the model and identify possible overfitting or underfitting.

Prediction Phase

For prediction, an input eye image is resized to 128 × 128 pixels, normalized, and passed through the trained CNN model.

The model produces a probability score between 0 and 1 using the Sigmoid activation function.

- Probability > 0.5 → Myopia Detected
- Probability ≤ 0.5 → Normal Eye

The predicted result can be displayed on the Raspberry Pi touchscreen as part of the portable eye screening device.

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APPENDIX

Appendix A: Hardware Components

Component	Specification	Purpose
Raspberry Pi 4 Model B	Quad-core ARM Cortex-A72 Processor	Main processing unit
Sony IMX296 Global Shutter Camera	1.6 MP Monochrome Sensor	Eye image acquisition
850 nm Infrared LEDs	High-intensity IR illumination	Non-mydriatic eye imaging
VL53L1X ToF Sensor	Time-of-Flight Distance Sensor	Working distance measurement
MOSFET Driver Circuit	Logic-Level MOSFET	Switching IR LED array
3.5-inch TFT Touchscreen	Raspberry Pi Compatible Display	User interface

Appendix B: Software and Development Tools

Software	Purpose
Python 3.x	Programming Language
TensorFlow	Deep Learning Framework
Keras	CNN Model Development
OpenCV	Image Processing
NumPy	Numerical Computation
Pandas	Dataset Management
Matplotlib	Performance Visualization
Scikit-learn	Data Splitting and Evaluation
KaggleHub	Dataset Download
Google Colab	Model Training Environment
Raspberry Pi OS	Embedded System Operating System

Appendix C: AI Model Parameters

Parameter	Value
Image Size	128 × 128 pixels
Batch Size	32
Epochs	10
Optimizer	Adam
Loss Function	Binary Crossentropy
Activation Function	ReLU
Output Activation	Sigmoid
Dropout Rate	0.3
Validation Split	20%

D: Dataset Information

Dataset Name: Myopia Image Dataset
Source: Kaggle
Classes:

- Myopia Images
- Normal Images

Image Format:

- JPG
- JPEG
- PNG

Image Size : 128 × 128 pixels

Appendix E: CNN Architecture Summary

Layer	Configuration
Input Layer	128 × 128 × 3
Convolution Layer 1	32 Filters (3 × 3), ReLU
Max Pooling Layer	2 × 2
Convolution Layer 2	64 Filters (3 × 3), ReLU
Max Pooling Layer	2 × 2
Convolution Layer 3	128 Filters (3 × 3), ReLU
Max Pooling Layer	2 × 2
Flatten Layer	Feature Vector Generation
Dense Layer	128 Neurons, ReLU
Dropout Layer	0.3
Output Layer	1 Neuron, Sigmoid

Appendix F: Model Training Workflow

1. Download dataset using KaggleHub.
2. Load image paths into a Pandas DataFrame.
3. Assign numerical labels (Myopia = 1, Normal = 0).
4. Split the dataset into training and validation sets.
5. Resize images to 128 × 128 pixels.
6. Normalize pixel values.
7. Train the CNN model.
8. Save the best-performing model.
9. Evaluate model accuracy and loss.
10. Perform prediction on new eye images.

Appendix G: Sample Prediction Output

Input Image	Predicted Class	Output
Eye Image 1	Myopia	Myopia Detected
Eye Image 2	Normal	Normal Eye

ANNEXURE

Annexure A: Software Environment

The AI model and image processing algorithms were developed using the following software tools:

Software/Tool	Version	Purpose
Python	3.x	Programming Language
Google Colab	Cloud IDE	Model development and training
TensorFlow	2.x	Deep Learning Framework
Keras	2.x	CNN Model Development
OpenCV	4.x	Image Processing
Pandas	Latest	Dataset Handling
NumPy	Latest	Numerical Computation
Matplotlib	Latest	Performance Visualization
Scikit-learn	Latest	Dataset Splitting
KaggleHub	Latest	Dataset Download

Annexure B: AI Model Implementation

The Convolutional Neural Network (CNN) was implemented using TensorFlow and Keras. The model consists of three convolutional layers followed by max-pooling layers, a fully connected dense layer, and a sigmoid output layer for binary classification.

Model Configuration

Parameter	Value
Input Image Size	128 × 128
Optimizer	Adam
Loss Function	Binary Crossentropy
Batch Size	32
Epochs	10
Activation Function	ReLU
Output Activation	Sigmoid

Annexure C: Dataset Preparation Code

```
import kagglehub

path = kagglehub.dataset_download("kellysanderson/myopia-image-dataset")

print(path)
```

Annexure D: Data Preprocessing Code

```
train_df, val_df = train_test_split(
    df,
    test_size=0.2,
    random_state=42
)
```

Annexure E: CNN Model Code (Excerpt)

```
model = Sequential([
    Conv2D(32,(3,3),activation='relu'),
    MaxPooling2D(2,2),
```

```
    Conv2D(64,(3,3),activation='relu'),
    MaxPooling2D(2,2),

    Conv2D(128,(3,3),activation='relu'),
    MaxPooling2D(2,2),

    Flatten(),

    Dense(128,activation='relu'),

    Dropout(0.3),

    Dense(1,activation='sigmoid')
])
```

Annexure F: Model Training Code

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=10
)
```

Annexure G: Prediction Code

```
img = image.load_img(img_path,target_size=(128,128))
img_array = image.img_to_array(img)/255.0
img_array = np.expand_dims(img_array,axis=0)

prediction = model.predict(img_array)
```